SHANGHAI JIAO TONG UNIVERSITY

Impact of Seat Occupying on Canteen Efficiency and Optimization: A Python-Based Study

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May. 31st, 2022

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overview

1 Background

2 Model Building

3 Questionaire Survey

4 Simulation & Analysis

5 Machine Learning

6 Conclusion

May. 31st, 2022

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A Python-Based Study

PART ONE

Background

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occupy a seat or not?

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No?

Perhaps...

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Probably...

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occupy a seat or not?

one's own convenience

other people's feelings

How to
strike a balance?

behavioral economics?

game theory?

not in our work!

How to
strike a balance?

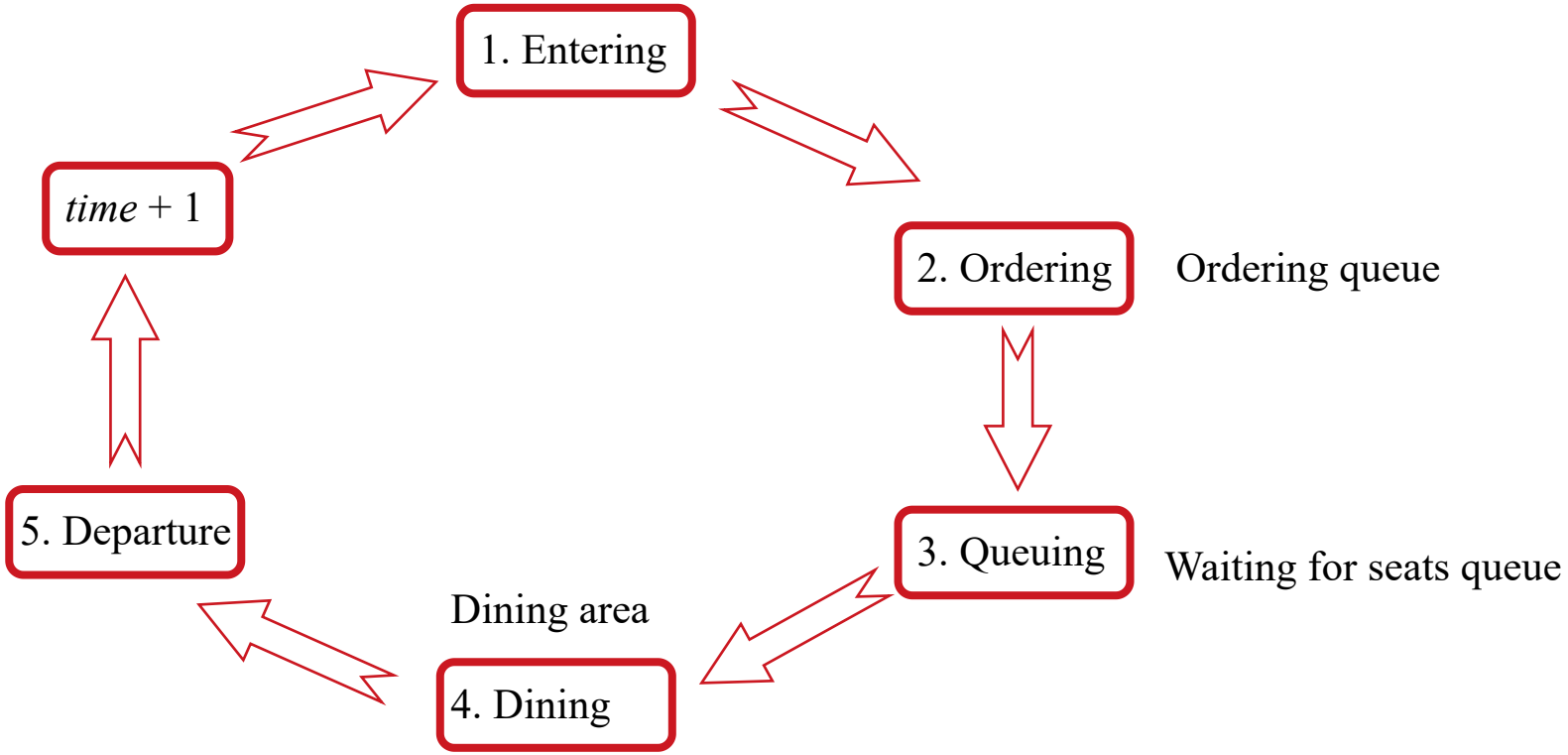
influence?
optimization?

we are interested in!

PART TWO

Model Building

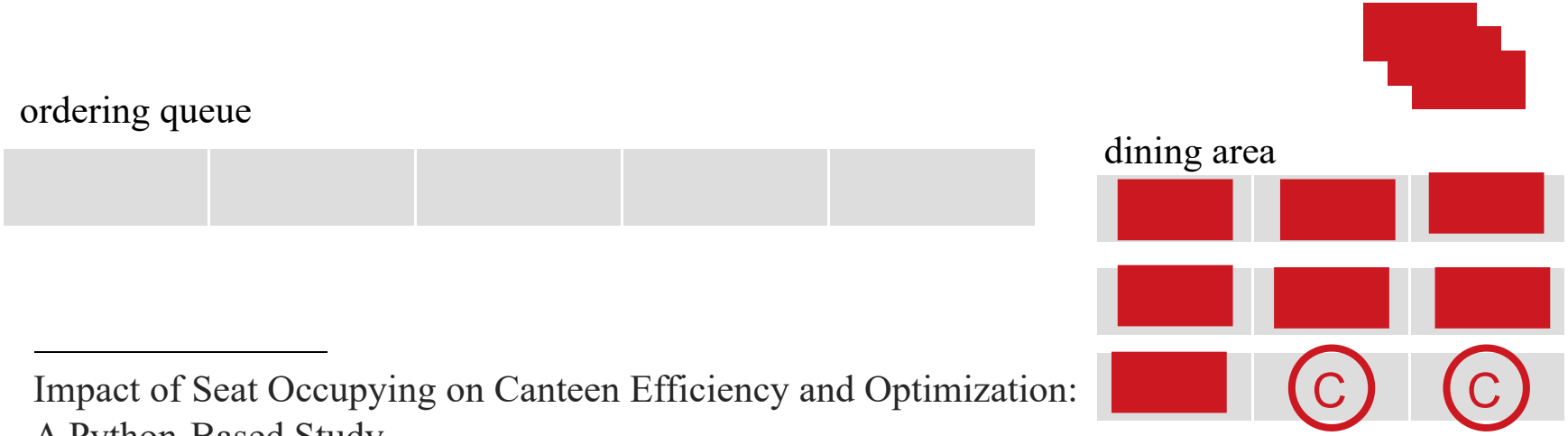
I. Base model


1. Entering
2. Ordering
Ordering queue
3. Queuing
Waiting for seats queue
Dining area
4. Dining
5. Departure
time + 1

1. Entering

Model 1:

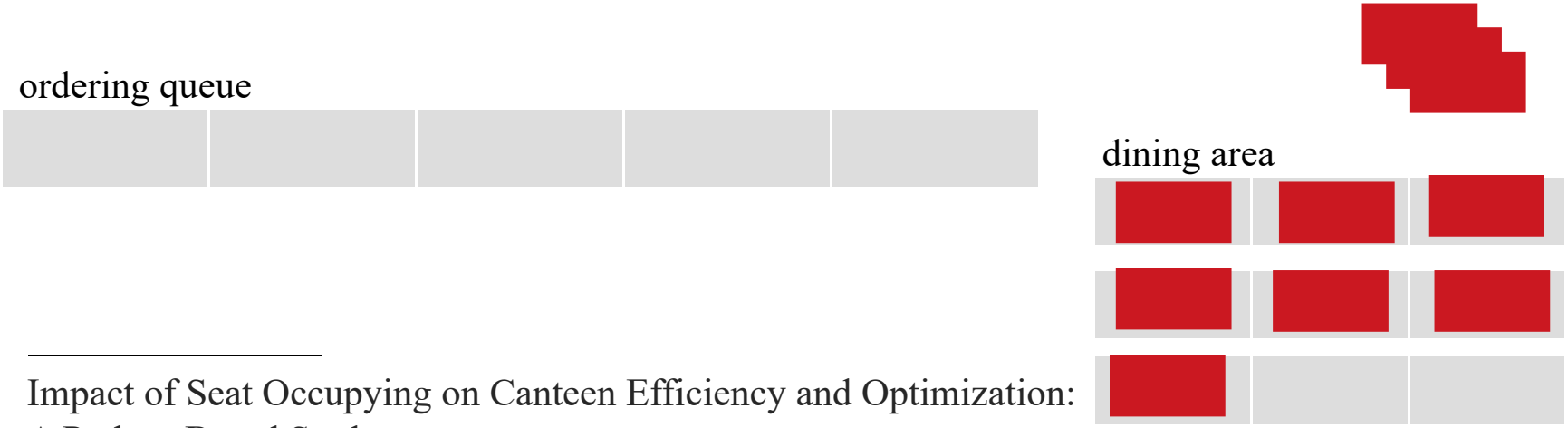
As soon as one student went into the canteen, if there's a empty seat, then occupy it. Otherwise go to the ordering queue.



1. Entering

Model 2:

As soon as one student went into the canteen, go directly to the ordering queue.



2. Queuing

Model 1&2:

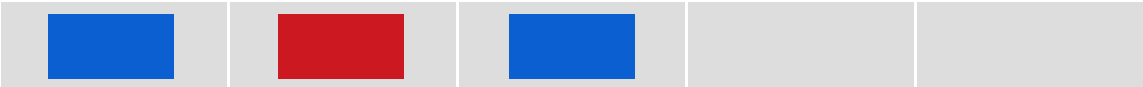
For each student queuing for seats, if there's an empty seat, then go and take it.

3. Ordering

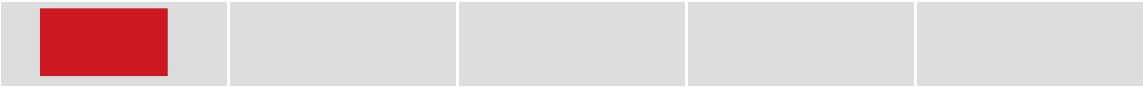
Model 1:

For each student in ordering line (fewer or equal the number a window can take in 1 min), if he/she has occupied a seat, then go have meal, otherwise go to ordering line.

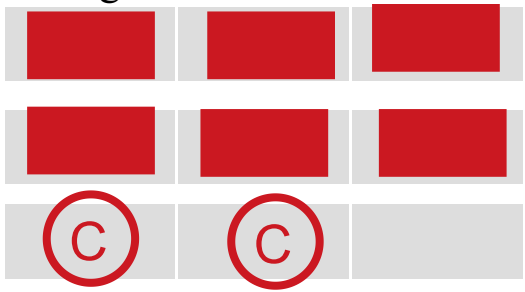
odering queue



waiting for seats queue



dining area



3. Ordering

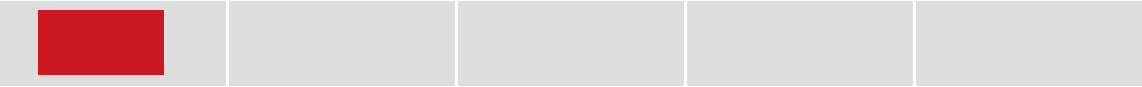
Model 2:

As soon as one student went into the canteen, if there's a empty seat, then occupy it.

Otherwise go to the ordering queue.

odering queue

waiting for seats queue



4. Departure

Model 1&2:

For each student in dining area, if he/she has finished eating, then leave the canteen and the number of empty seats plus one.

II. Problems & Solutions

Problem: The total dining times are almost the same.

Reason: Students in seat waiting line always get the priority to take seats

1. Students arrives in a total random way

2. Not all students grab a seat when entering the canteen

3. Students in seat waiting line shouldn't always have the priority to take a seat

4. Different window have different window attractiveness.

1. Seat getting priority

Add another parameter that decide the proportion of seats the students in the seat waiting line can get.

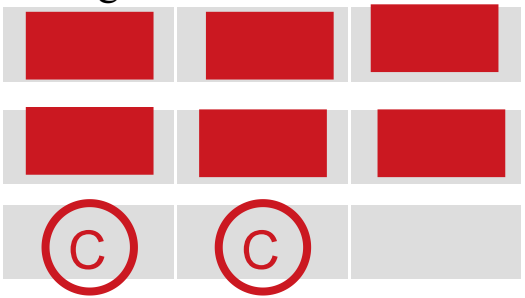
students arriving

given proportion=0.3

odering queue

seat waiting queue

dining area



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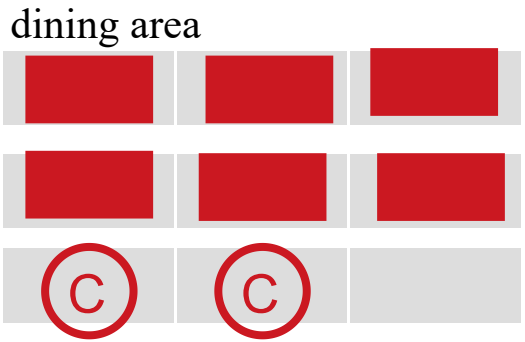
1. Seat getting priority

Using (the number of people actually eating) / (seats taken) be the evaluation of the model

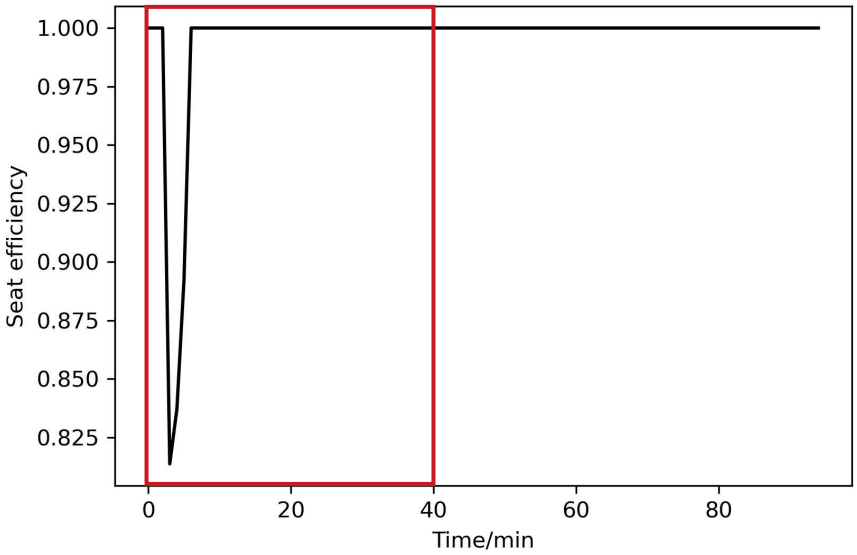


$$\frac{66}{86} = 0.75$$

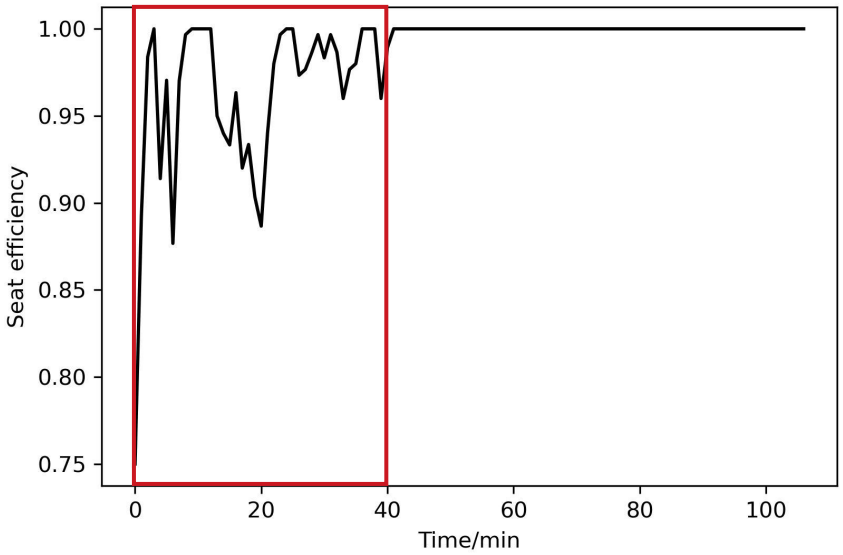
1. Seat getting priority

Using (the number of people actually eating) / (seats taken) be the evaluation of the model

Old model



New model (*seat-get-rate* = 0.3)



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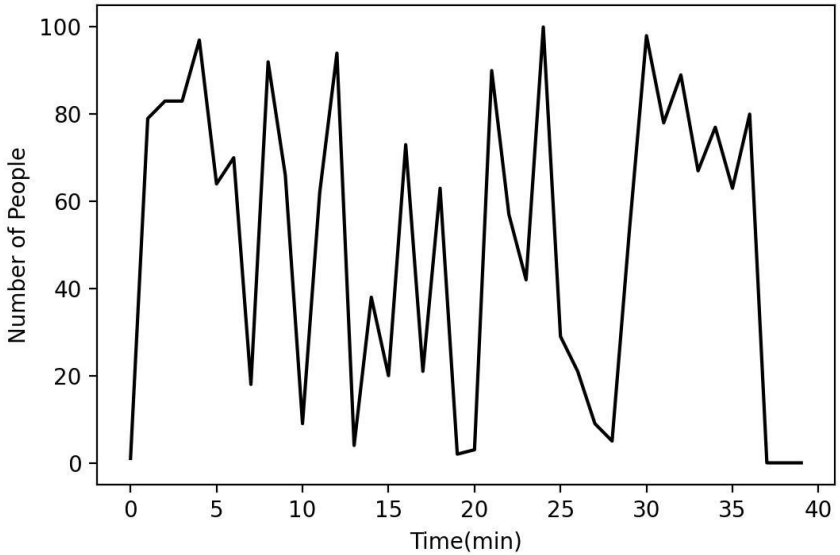
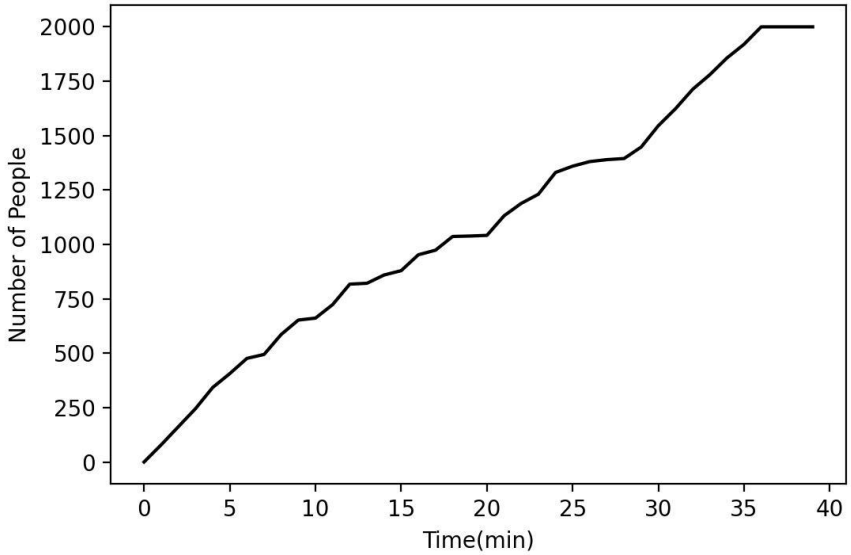
2. Arriving model

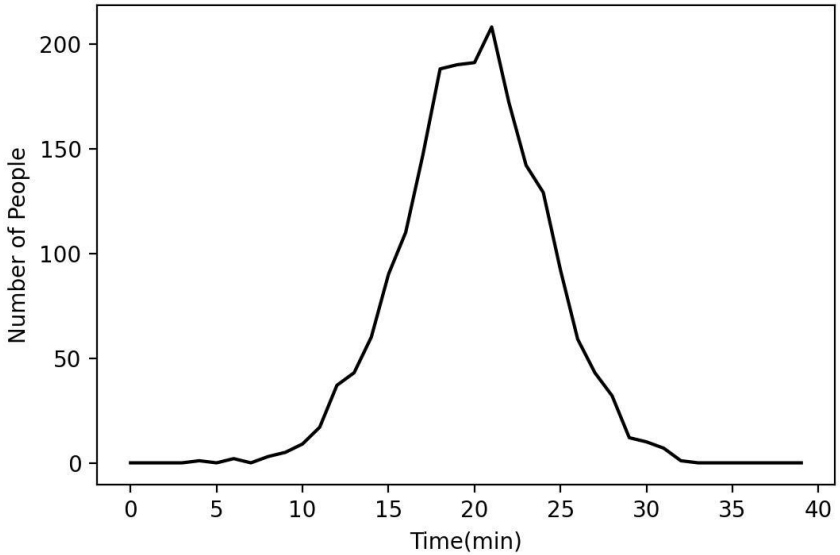
Old model: Totally random, we don't limit the overall feature.

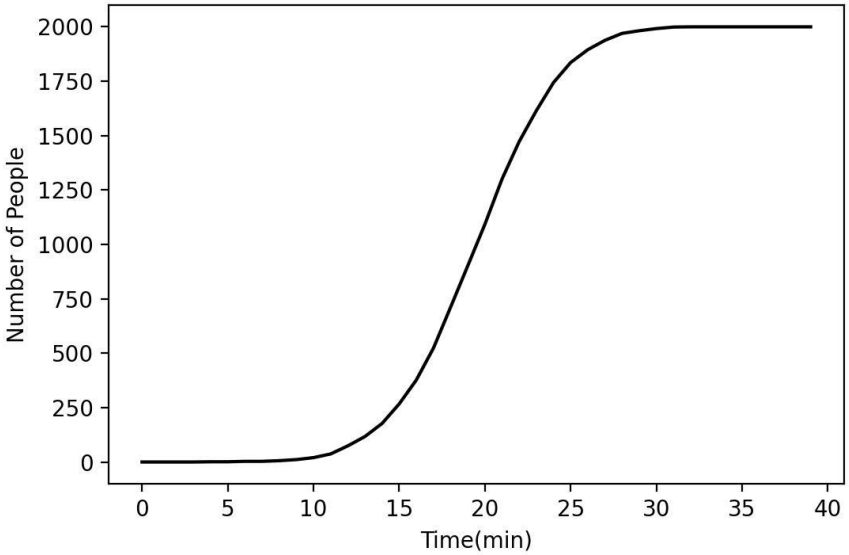




2. Arriving model

New model: Using normal distribution to create a peak period





2. Arriving model

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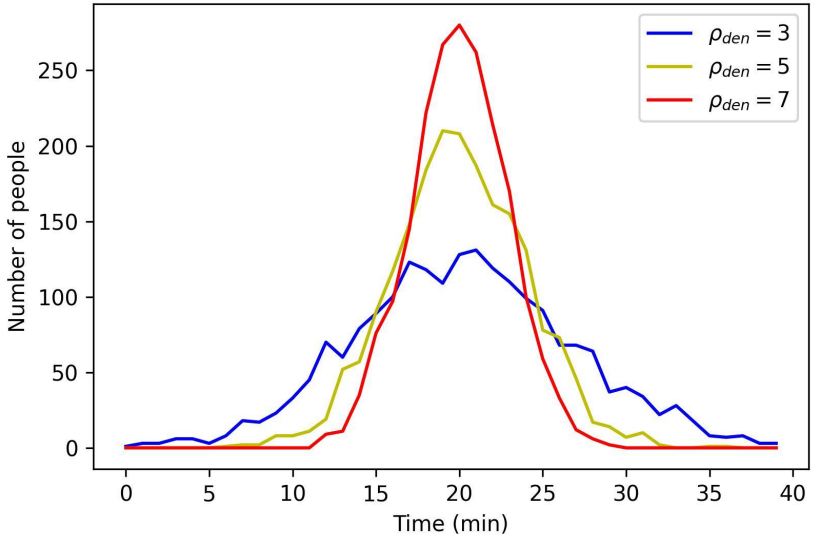
A new parameter

Number of student arriving in each minute (diffenrent density)



Suppose the total number of students is n

Then $\sigma = \dfrac{n}{2\rho_{den}}$

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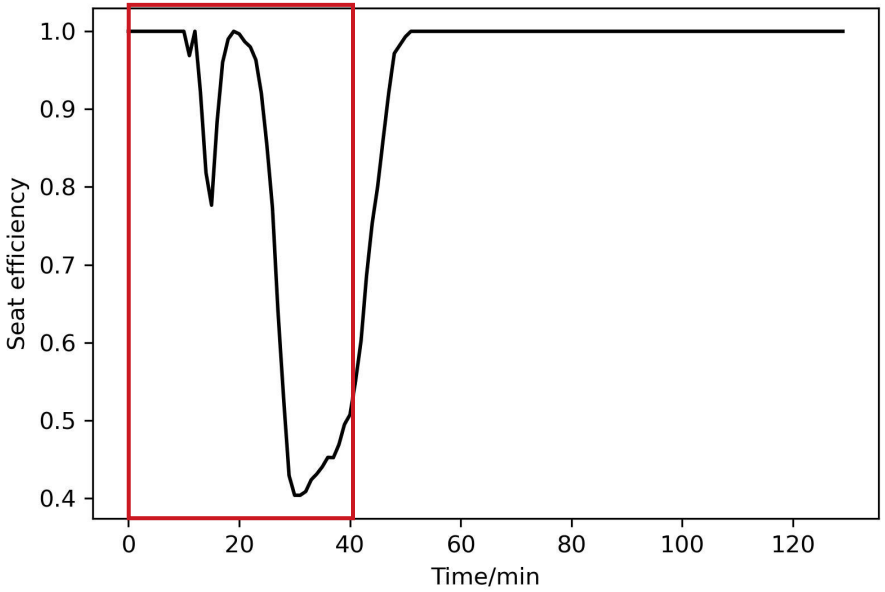
2. Arriving model

Former concern on seat priority

Normal distribution model efficiency (*seat-get-rate* = 0.3)



3. The occupy rate

A new parameter deciding the possibility of a student who just arrived taking a seat or not.

given rate=0.5

random number(0, 1): 0.38

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4. Window attractiveness

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A new parameter deciding the possibility of a student went to each of the window

win1 0.1-0.4	win2 0.4-0.5	win3 0.5-0.8	win4 0.9-1.0

students arriving

random number(0, 1): 0.45

PART THREE

Questionnaire Survey

data of a real canteen

collected through

a survey

I. Data to be collected

How many people would...

go to canteen
instead of orderimg takeout

occupy a seat

II. Questionnaire

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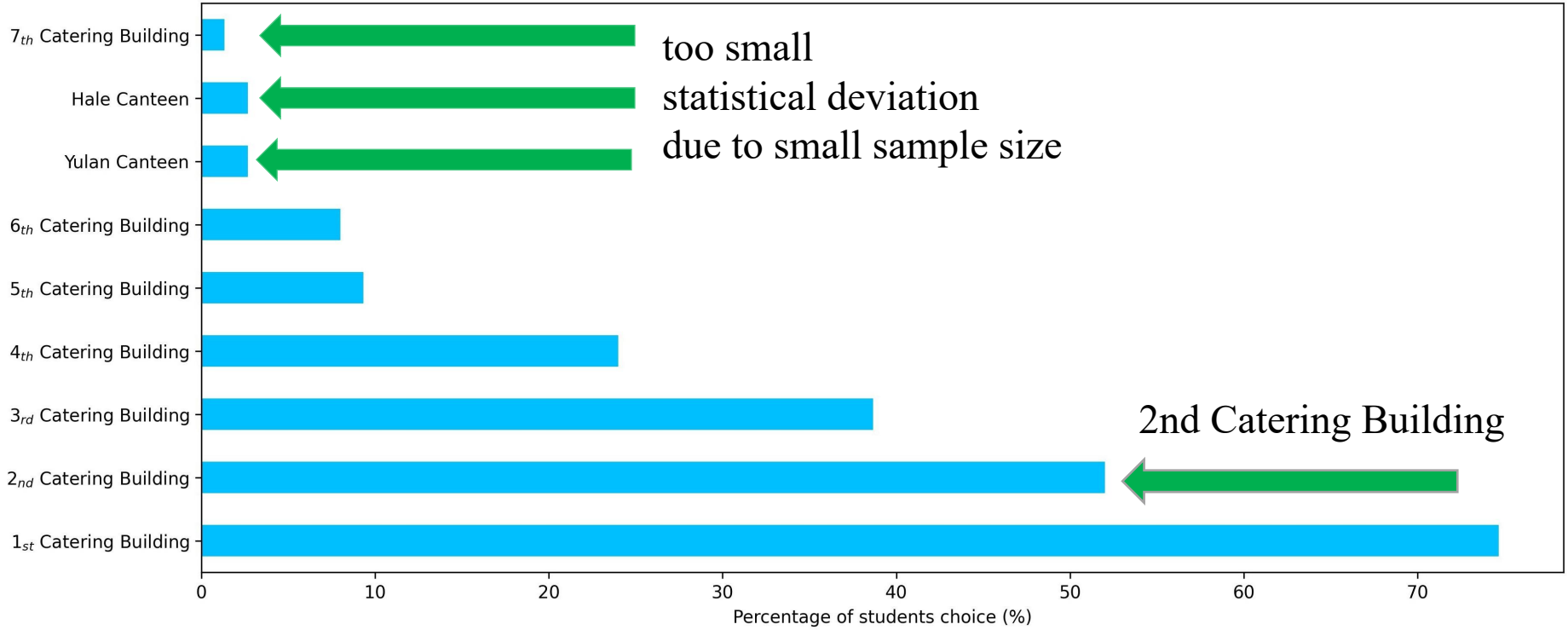
Questionnaire

time	subject	items	number
in late March 2022	undergraduates in Minhang campus, SJTU	4 closed multiple-choice questions	75 valid ones

III. Bar chart



IV. Data process

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项目\频率	总是	经常	偶尔	极少或从未
在食堂就餐频率	49.33%	40%	5.33%	5.33%
对应百分频率	95%	65%	35%	5%
加权平均结果	0.74729			
闵行校区在校生	34000			
食堂就餐人数（教职工3000）	28407			

项目\频率	经常	偶尔	很少	从未
目睹占座行为	56%	22.67%	16%	5.33%
本人占座行为	15%	28%	14.67%	42.67%
对应频率	75%	50%	25%	0
目睹-本人加权平均	0.54	0.23	0.16	0.07
最终频率加权平均	0.560			

最常去的食堂	总和213.34%	座位数（交我办）	人数-座位比	理论接纳人数	全校区用餐人数
第二餐饮大楼	74.67%	2020	4.9	9942	28407
第一餐饮大楼	52%	1380	5.0	6923	
玉兰苑	38.67%	216	23.8	5149	
第四餐饮大楼	24%	1354	2.4	3195	
第三餐饮大楼	9.33%	1184	1.0	1242	
哈乐餐厅	8%	367	2.9	1065	
第六餐饮大楼	2.67%	608	0.6	355	
第五餐饮大楼	2.67%	637	0.6	355	
第七餐饮大楼	1.33%	248	0.7	177	

the total diners

$34000 \times 0.74729 + 3000 \approx 28407$

take “usually” choice as an example

$56\% \times 0.95 + 15\% \times 0.15 \approx 0.54$

trivival process

seats number from “mysjtu”

V. Table

Variable name	Quantity	
Total number of students	9942	
people number in peak period	5965	arriving model
Total number of seats	2020	
Total arriving time (min)	140	arriving model
peak period	40	arriving model
Total number of windows	48	
Speed of eating (min)	12	not availble
One window can deal in 1 min	3	we set a number

data

put into

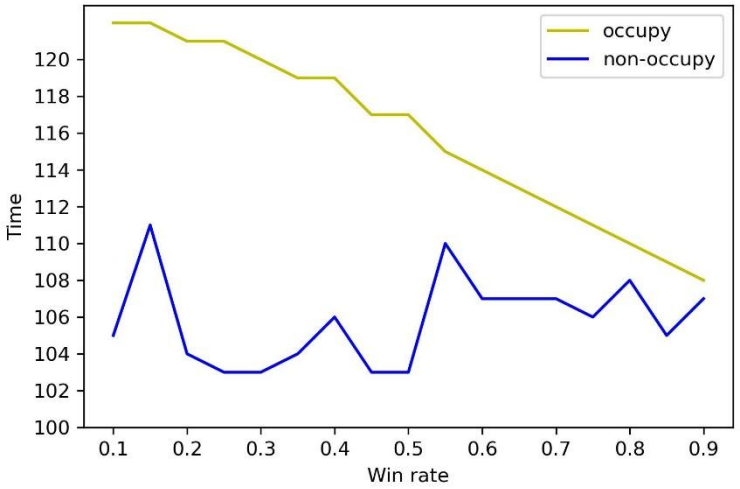
model

PART FOUR

Simulating and Analysis

How we analyze our data?

- Set our base data based on our questionaires
- Change one parameter one at a time
- Analyze its influence on total time consumption
- Use Python to visualize the result

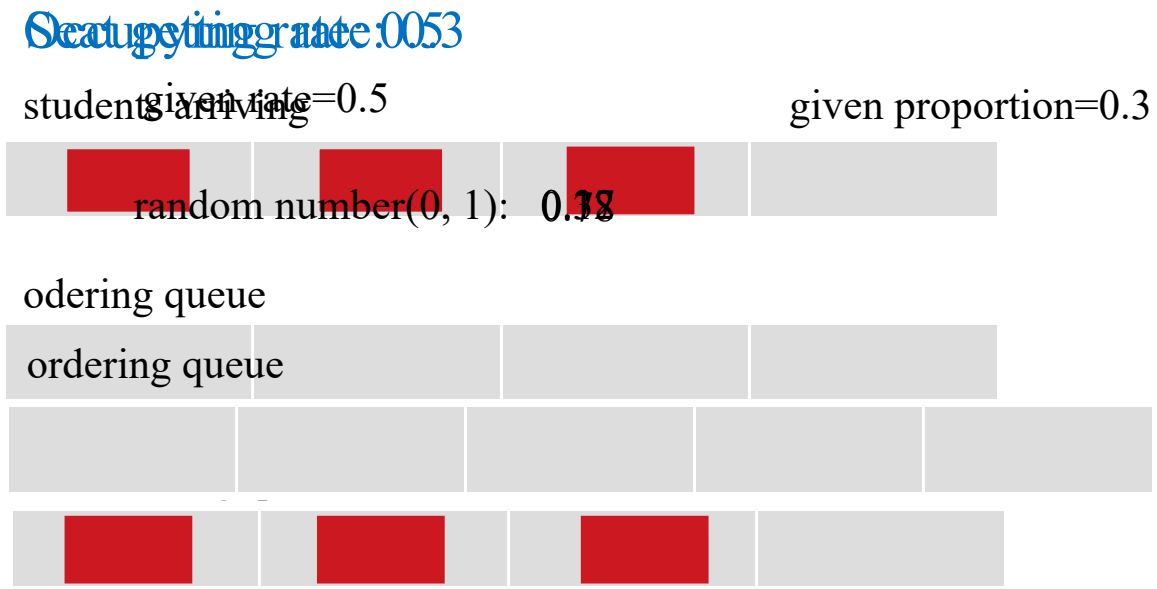


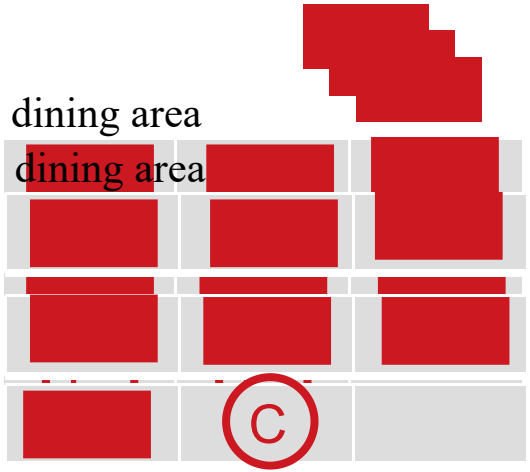
Std number	Seat number	Seat getting rate	Occupying rate	Window attributions	Eating speed
8000	3000	0,8 0.1, 0.2, 0.3 , … 0.9	0.6	…	3

Why in this way

- Save computing cost
- Close to real situation
- Help evaluate which parameter has the biggest impact on our model

I. Seat getting rate & Occupying rate





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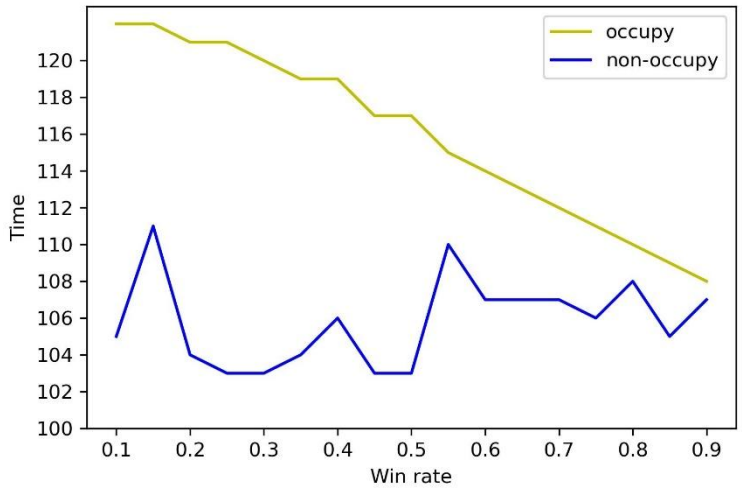
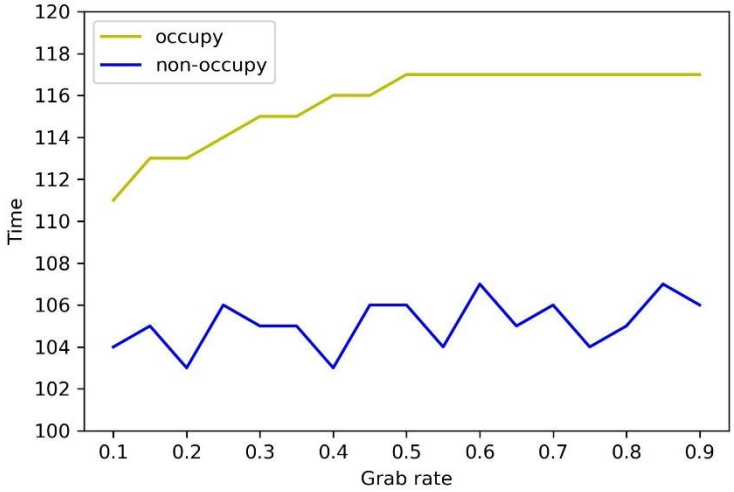
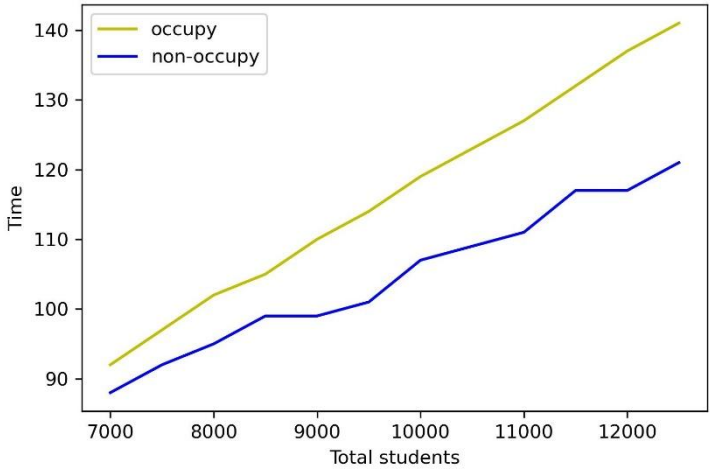
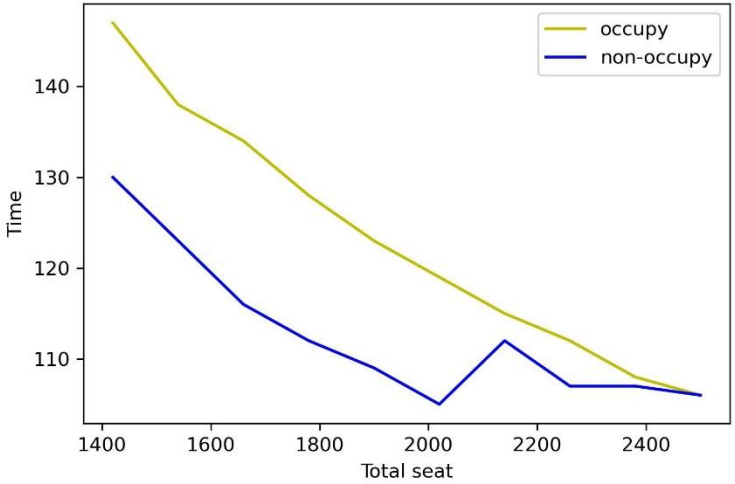
I. Seat getting rate & Occupying rate



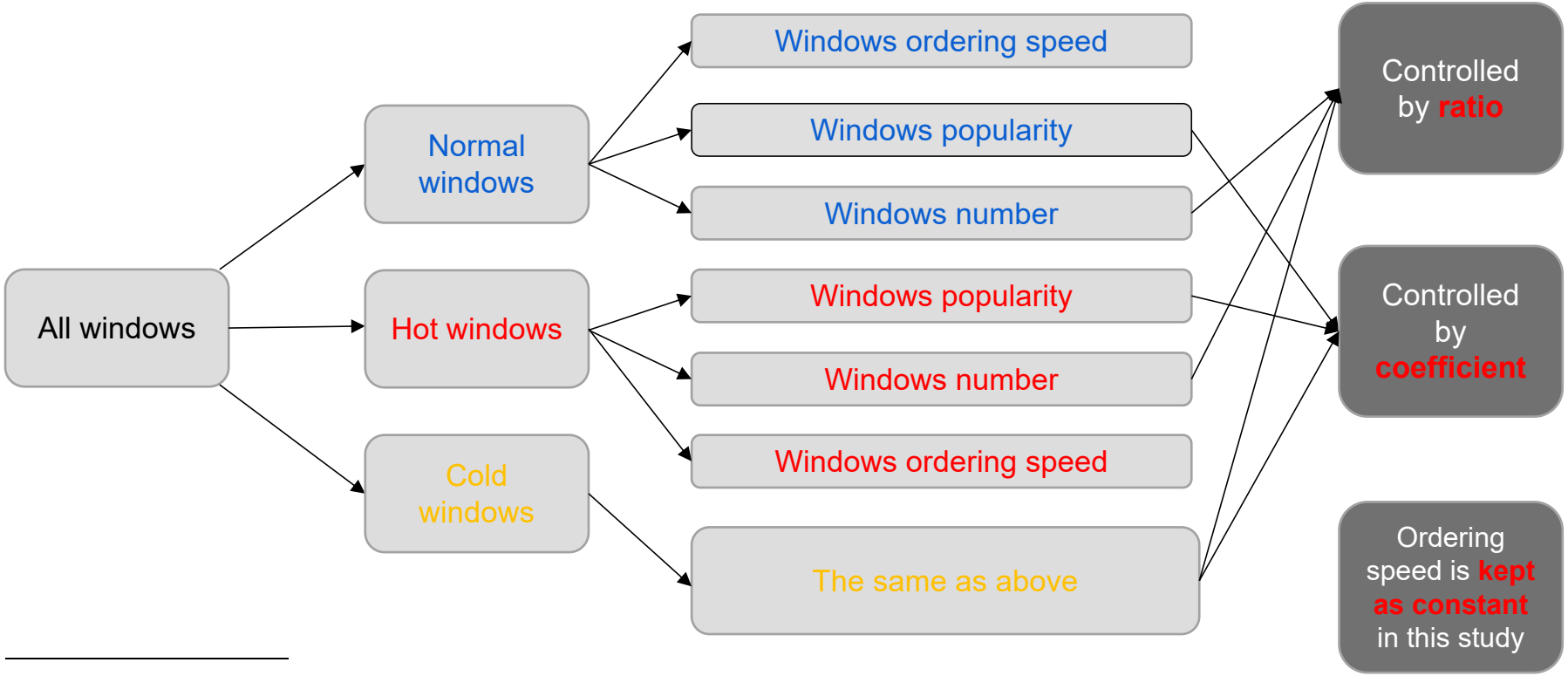


II. The number of students and seats





III. The influence of window attributions



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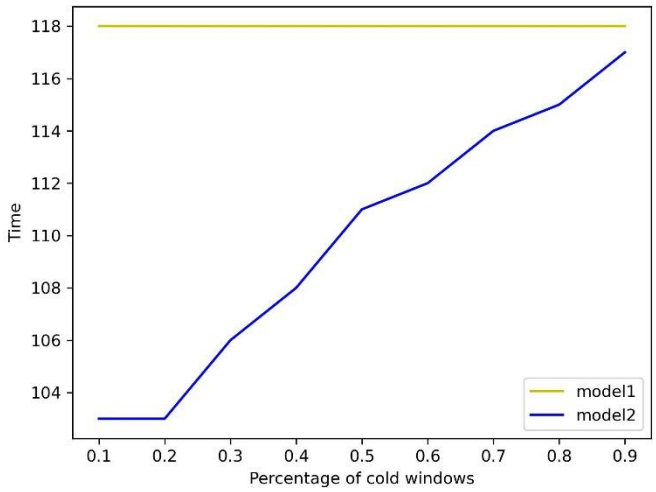

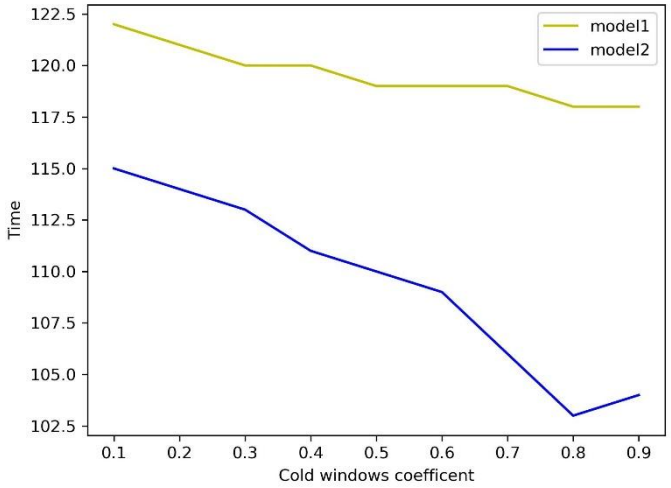

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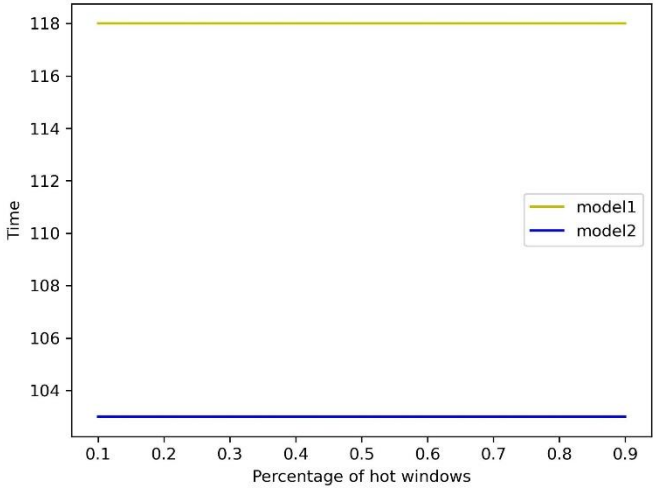
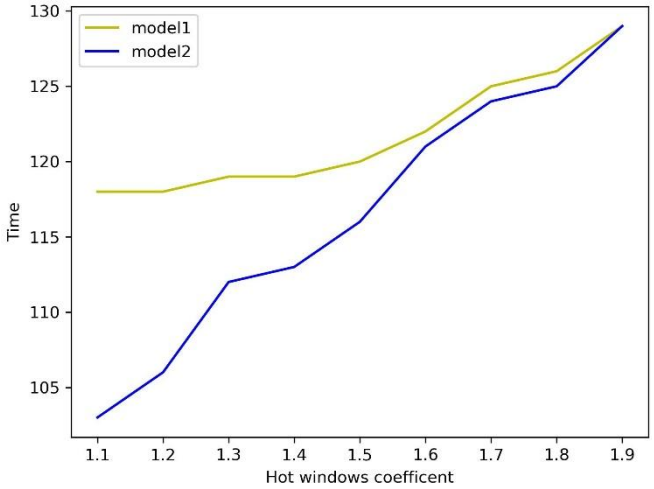
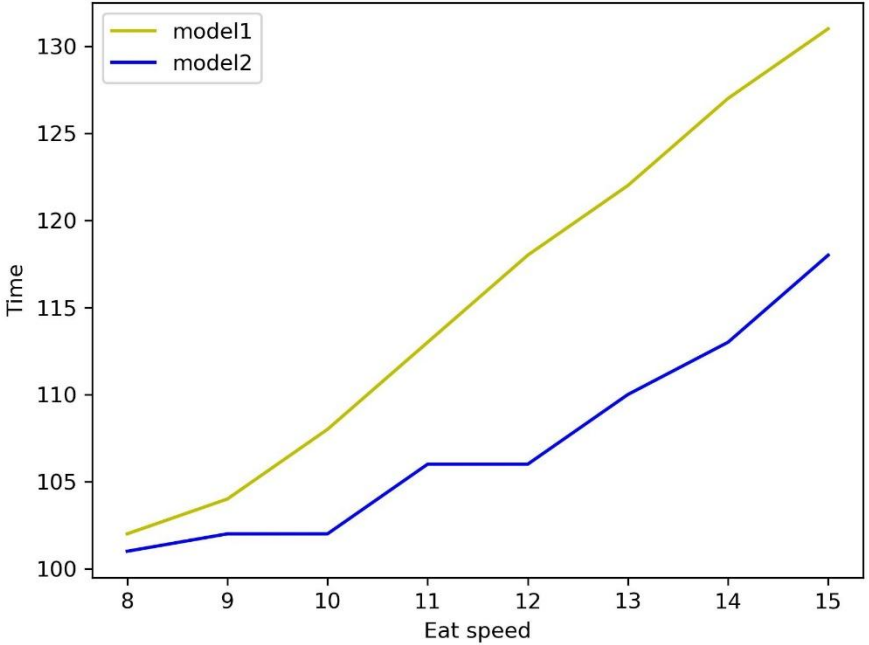




IV. Eating speed



PART FIVE

Machine learning

I. Data collecting

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Prepare the dataset

To estimate

Input

	total_time	stu_num	arrive_time	win_rate	grab_rate	eat_speed	seat_num	density
0	88	7000	60	0.4	0.4	10	1540	7
1	99	7000	60	0.4	0.4	13	1540	7
2	112	7000	60	0.4	0.4	16	1540	7
3	84	7000	60	0.4	0.4	10	1660	7
4	96	7000	60	0.4	0.4	13	1660	7
...	...	...	...	...	...	...	...	...
44923	115	13000	80	0.9	0.9	13	2260	3
44924	131	13000	80	0.9	0.9	16	2260	3
44925	108	13000	80	0.9	0.9	10	2380	3
44926	114	13000	80	0.9	0.9	13	2380	3
44927	127	13000	80	0.9	0.9	16	2380	3

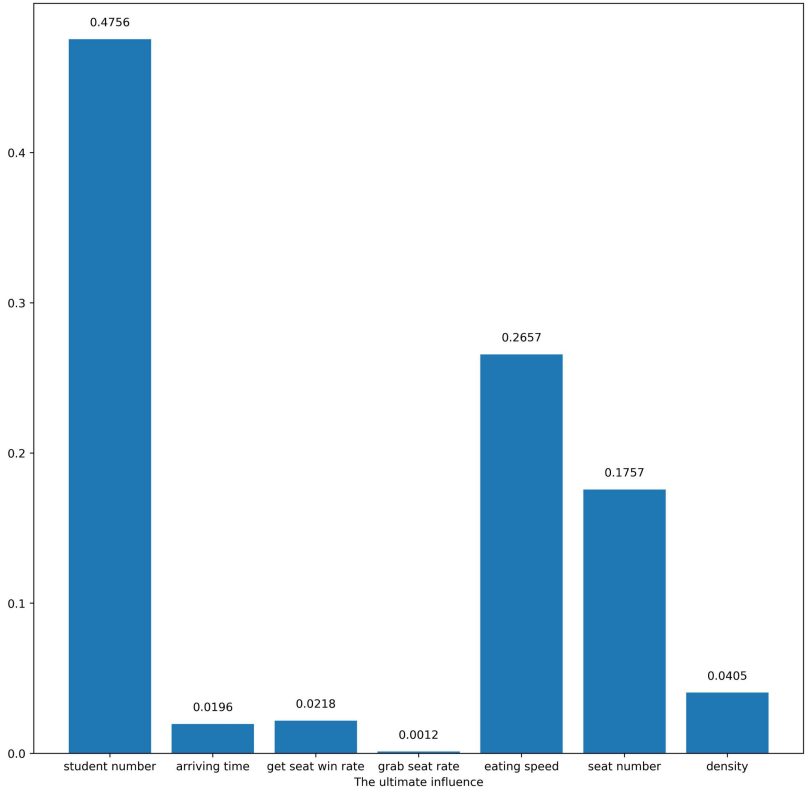
44928 rows × 8 columns

II. RandomForest model

Accuracy

Linear model: 92.97%

RandomForest model: 99.04%



III. Neural Network model

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Input:

student number

arriving time

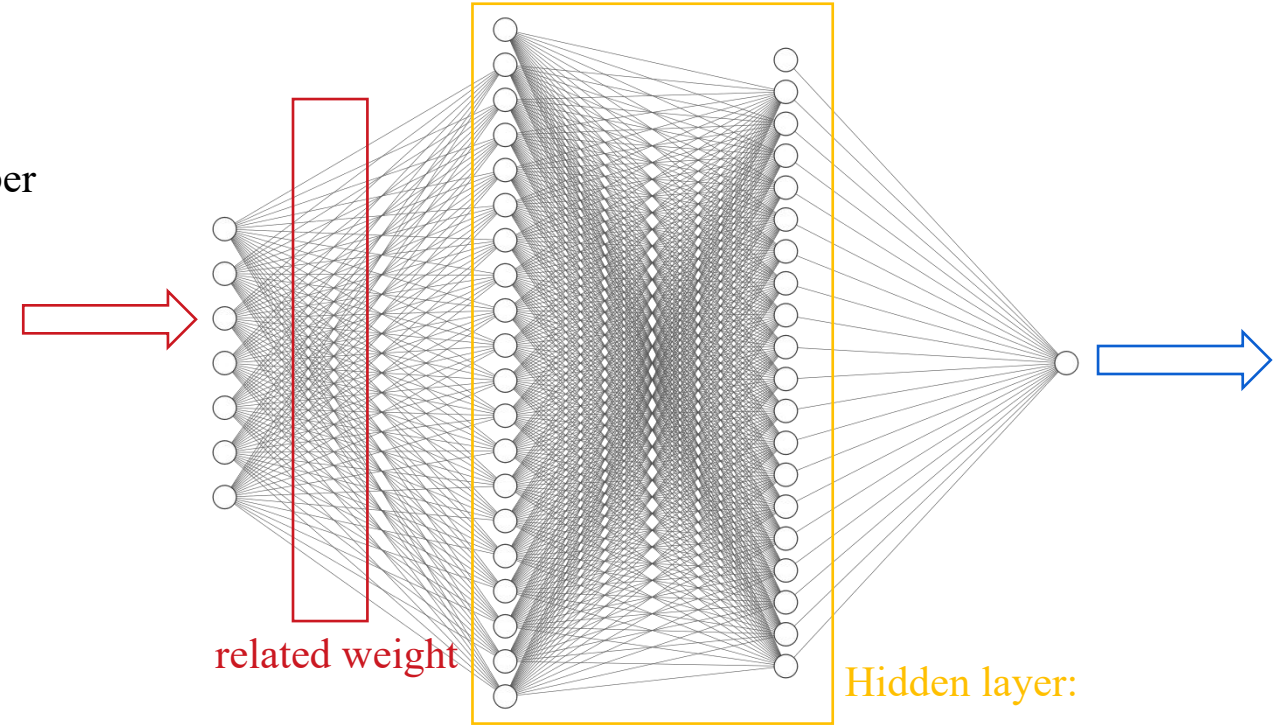
win-get rate

eating speed

grab rate

seat number

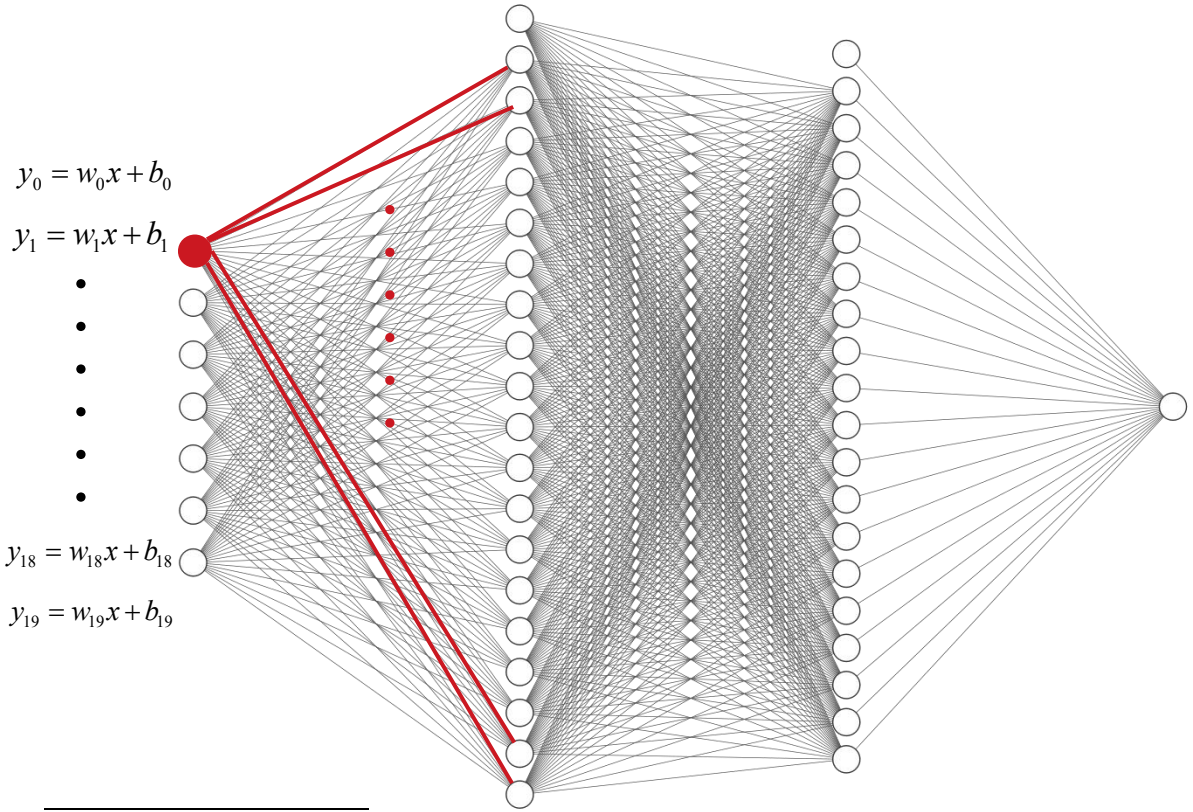
density



Output:

total time

III. Neural Network model



For each edge: $y = wx + b$

The final weight of each parameter

$$W = \sum_{i=0}^{19} w_i$$

IV. Comparison

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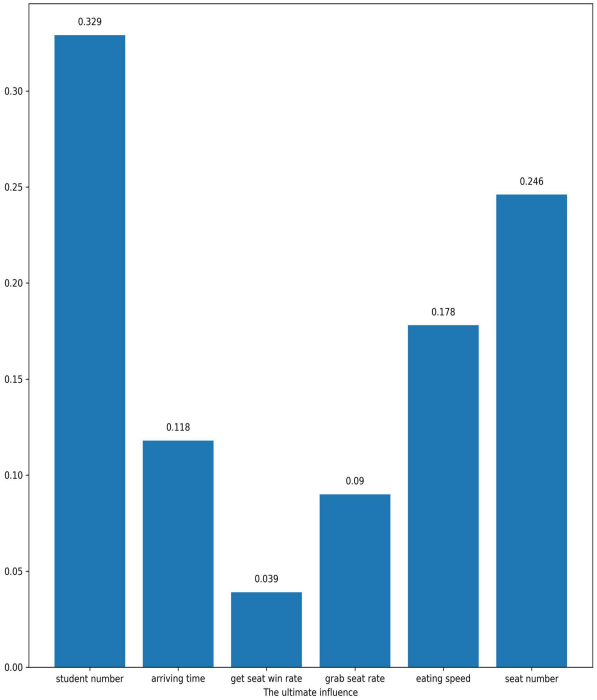
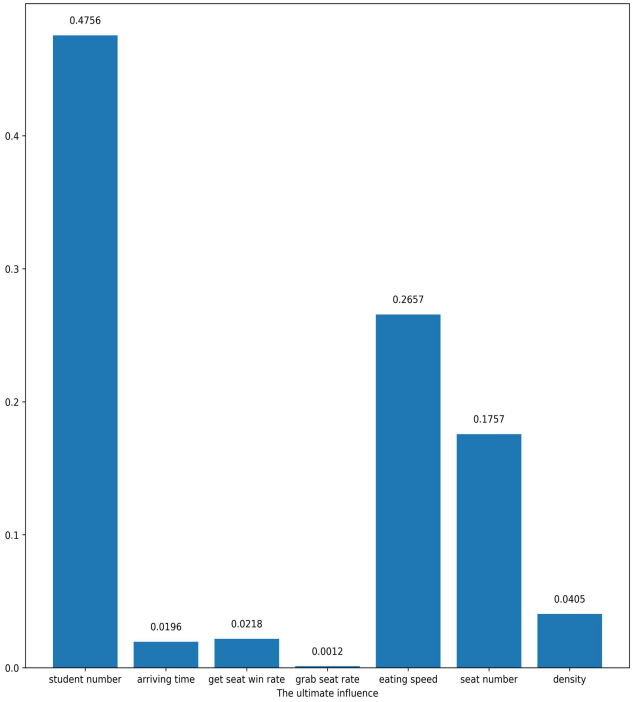
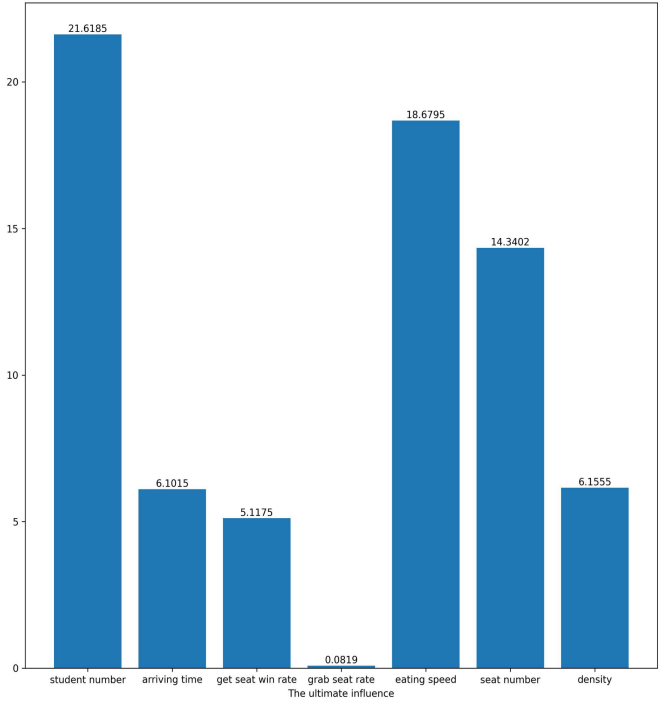


Own fomula



RandomForest model



Neural Network model

Impact of Seat Occupying on Canteen Efficiency and Optimization: A Python-Based Study

PART SIX

Conclusion

I. Focus

fomula model

- Set our base data based on our questionaires
- Change one parameter one at a time
- Analysis its influence on total time consumption

supplementary

RandomForest model

Neural Network model

Impact of Seat Occupying on Canteen Efficiency and Optimization:
A Python-Based Study

II. ranking list

1. student number

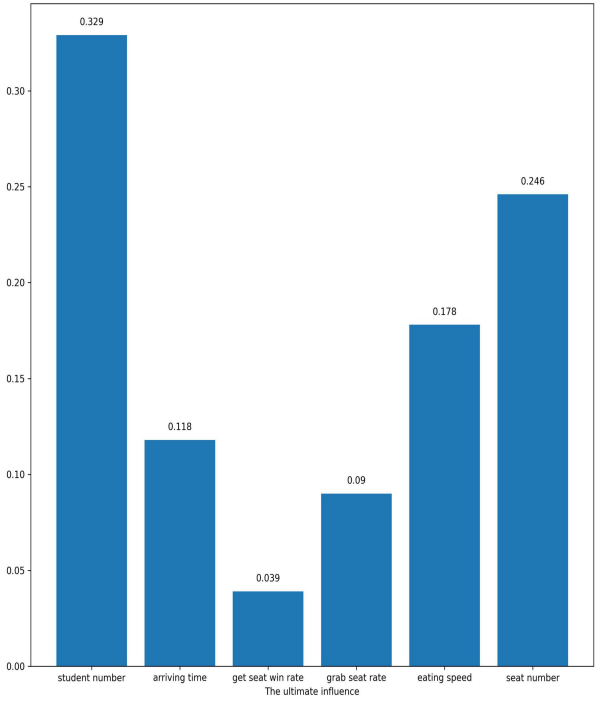
2. seat number

3. eating speed

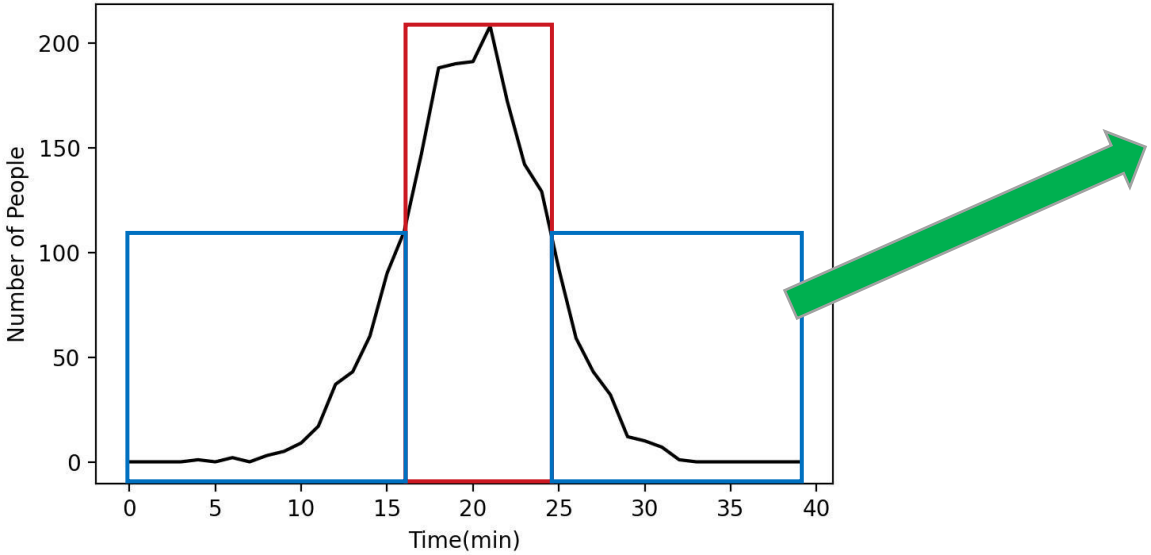
4. arriving time

5.grab seat rate

6.wait and get rate



III. Possible explanations



not necessary

little impact

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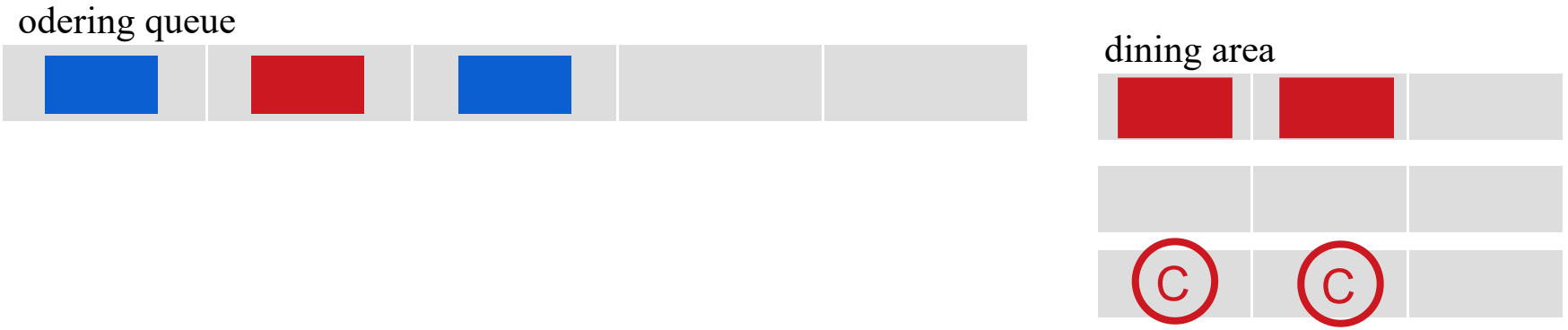
III. Possible explanations



IV. Optimization

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1. less and more balanced student number
2. more seats
3. finish in a proper time
4. avoid peak period
5. slogans and volunteers

ranking list

1. student number
2. seat number
3. eating speed
4. arriving time
5. grab seat rate
6. wait and get rate

IV. Limitations

I. model

1. daily details

2. friends and couples

II. questionnaire survey

1. small sample size

2.diversity of subject backgrounds

3. measurement errors

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Thank you for your attention!

May. 31st, 2022

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